DEEP LEARNING — PROJECT REVIEW

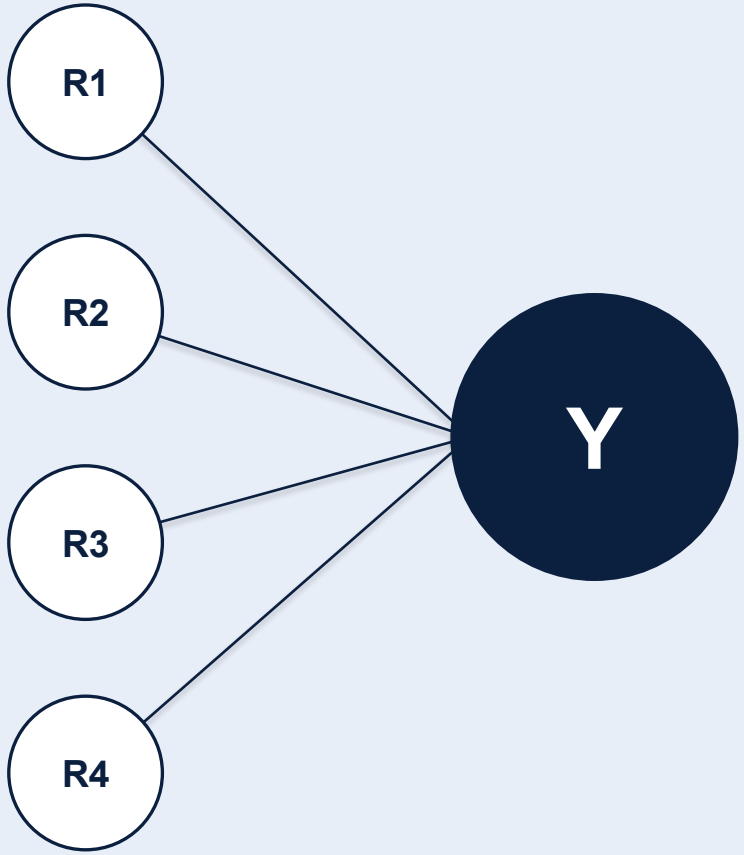
Knee Osteoarthritis Severity Grading from X-rays Using a From-Scratch ResNet Ensemble

Kellgren–Lawrence grade classification (G0–G4) with label cleaning, self-supervised pre-training and a four-seed ensemble

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Four ResNet seeds (R1–R4) → ensemble → KL grade Y

Project Implementation Workflow

Five stages from the raw X-ray to the final KL grade

01

Preprocessing & Augmentation

Grayscale 224×224 replicated to 3 channels and scaled to [−1, 1]; GPU augmentation (rotation, shift, zoom, flip, noise, erasing).

Output: 4,911 train / 867 validation / 4,008 test images

02

Label Cleaning with CleanLab

5-fold out-of-fold predictions flag suspect labels; flagged samples are down-weighted to 0.3× and never removed.

Output: 988 of 4,911 training labels down-weighted

03

Self-Supervised Pre-training

Rotation prediction (0° / 90° / 180° / 270°) for 15 epochs on the training images only; no labels, no outside data.

Output: Initial weights for every ResNet

04

Fine-tuning Four ResNet Seeds

80 epochs with class-weighted focal loss and ordinal soft labels; AdamW, cosine schedule, weight EMA of 0.998.

Output: Four models, validation κ 0.815 – 0.823

05

Ensemble & Decoding

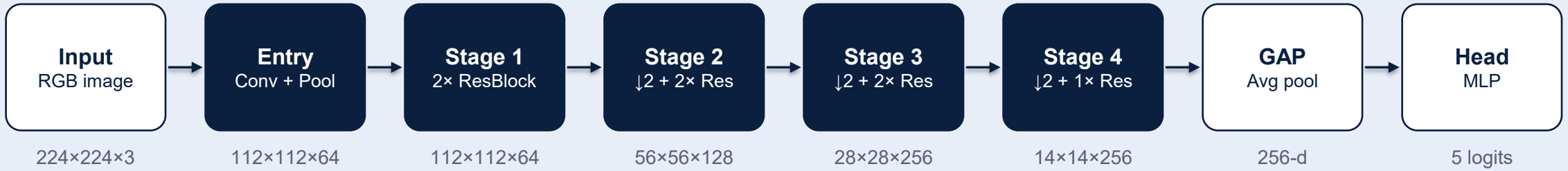
κ-weighted average with horizontal-flip test-time augmentation; per-class bias tuned on the validation set only.

Output: 65.7 % accuracy, quadratic κ 0.813

ResNet Architecture

A 5.31 M-parameter residual network trained from random initialisation

1 DATA FLOW



2 DESIGN CHOICES

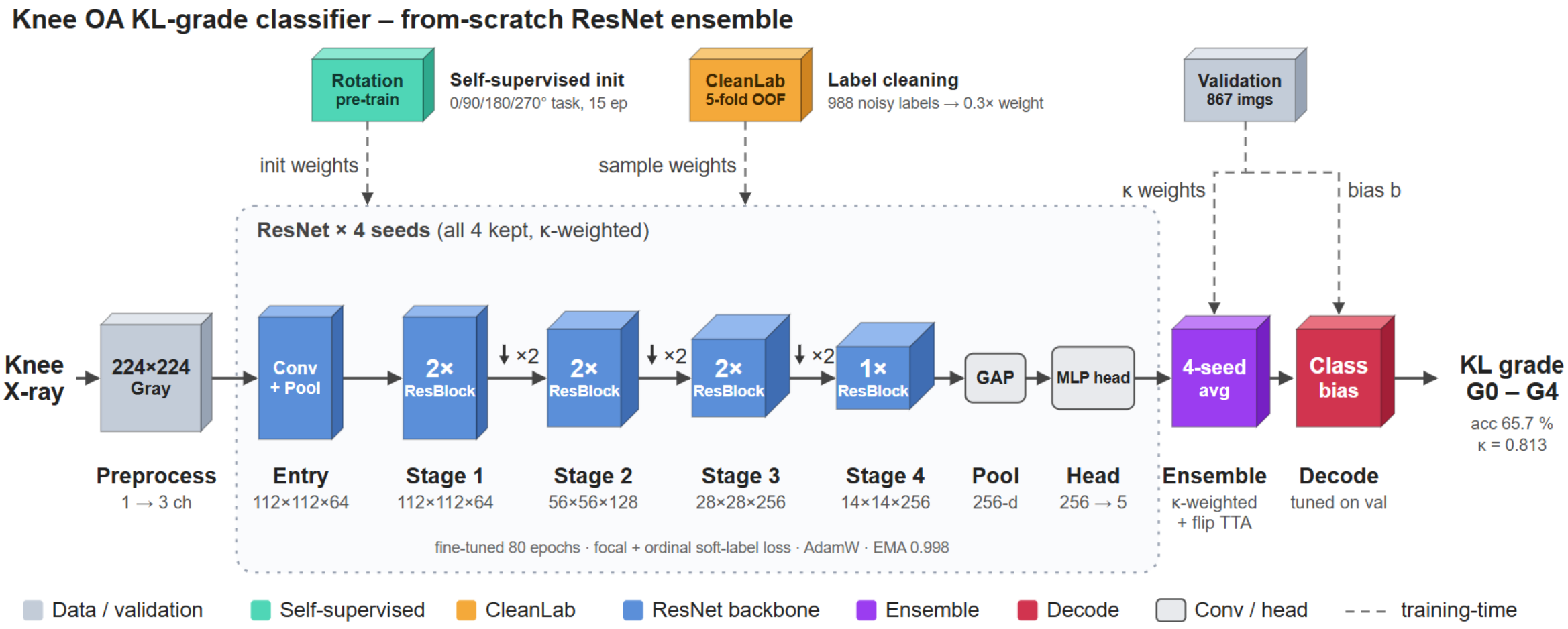
- **From scratch:** random initialisation, no pretrained weights
- **Four stages:** 64 → 128 → 256 → 256 channels
- **Overall stride 16:** 224×224 input reduced to a 14×14 map
- **Downsampling:** stride-2 convolutions instead of pooling
- **Compact:** about 25 minutes per model on a Kaggle T4

3 PARAMETERS BY COMPONENT

Component	Output	Parameters	Share
Entry	112×112×64	1,856	< 0.1 %
Stage 1	112×112×64	147,968	2.8 %
Stage 2	56×56×128	664,832	12.5 %
Stage 3	28×28×256	2,656,768	50.0 %
Stage 4	14×14×256	1,771,008	33.4 %
Head	5 logits	67,333	1.3 %
Total		**5,309,765**	**100 %**

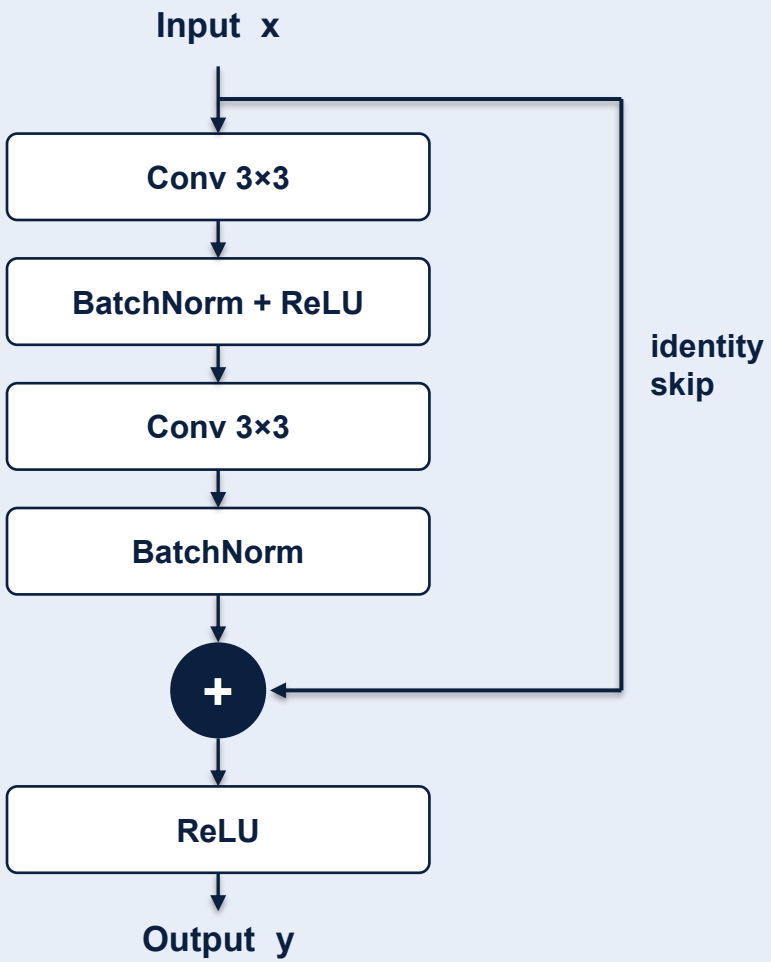
Model Architecture Overview

End-to-end pipeline: self-supervised init, label cleaning, four ResNet seeds, ensemble and decoding



Components: Residual Block & Downsampling

The two building blocks that make up Stages 1 to 4



1 RESIDUAL BLOCK

Two 3×3 convolutions with batch normalisation. The input is added back before the final ReLU: add, then activate.

$$y = \mathrm{ReLU}(\,\mathrm{BN}(\mathrm{Conv}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(x))))) + x\,)$$

2 WHY IT HELPS

The identity path gives gradients a direct route to early layers, which makes a four-stage network trainable from scratch.

3 DOWNSAMPLING BETWEEN STAGES

A stride-2 3×3 Conv-BN-ReLU halves the resolution and widens the channels (64 → 128 → 256), replacing pooling.

74 K	parameters in a ResBlock at 64 channels
1.18 M	parameters in a ResBlock at 256 channels

Components: Entry, Pooling & Classification Head

How the image is lifted into features and turned into a grade

1 ENTRY BLOCK

- Conv 3×3, 3 → 64
- BatchNorm + ReLU
- MaxPool 2×2

Lifts the 3-channel input to 64 feature maps and halves the resolution from 224 to 112.

1,856 parameters

2 GLOBAL AVERAGE POOLING

- 14×14×256 feature map
- Adaptive average pool
- 256-d vector

Averages each channel over space into a fixed-length descriptor that tolerates small shifts of the joint.

0 parameters

3 CLASSIFICATION HEAD

- Linear 256 → 256
- BatchNorm + ReLU
- Dropout 0.4
- Linear 256 → 5

Maps the descriptor to five KL-grade logits; dropout limits over-fitting on the small training set.

67,333 parameters

4 WHERE THE 5.31 M PARAMETERS SIT



Entry < 0.1 % Stage 1 2.8 % Stage 2 12.5 % Stage 3 50.0 % Stage 4 33.4 % Head 1.3 % (left to right)

Training Strategy

Loss, optimisation and inference choices, all tuned on the validation set

1 LOSS FUNCTION

- **Focal loss** (γ = 2) with class weights for the rare G4
- **Ordinal soft labels:** 15 % of the label mass goes to neighbouring grades
- **Ordinal MSE term** ramped up over 15 epochs (weight 0.05)
- **CleanLab weights:** suspect labels count 0.3×

2 OPTIMISATION

- **AdamW:** lr 1e-3, weight decay 2e-2 (not on BatchNorm or bias)
- **Schedule:** 3-epoch warm-up, then per-step cosine to 0
- **Weight EMA 0.998;** checkpoint chosen by validation κ
- **Mixed precision** (fp16) on a Kaggle T4

3 AUGMENTATION (ON THE GPU)

- Rotation ±15°, translation, zoom, horizontal flip
- Gamma, brightness and contrast jitter
- Gaussian noise and random erasing

4 INFERENCE

- **Flip test-time augmentation** on every model
- **κ-weighted average** of the four seeds
- **Per-class bias** tuned on validation macro-F1

Nothing is selected or tuned on the test images.

Comparison of the Three Architectures

Same data, same training recipe, four seeds each; scored on all 4,008 test images

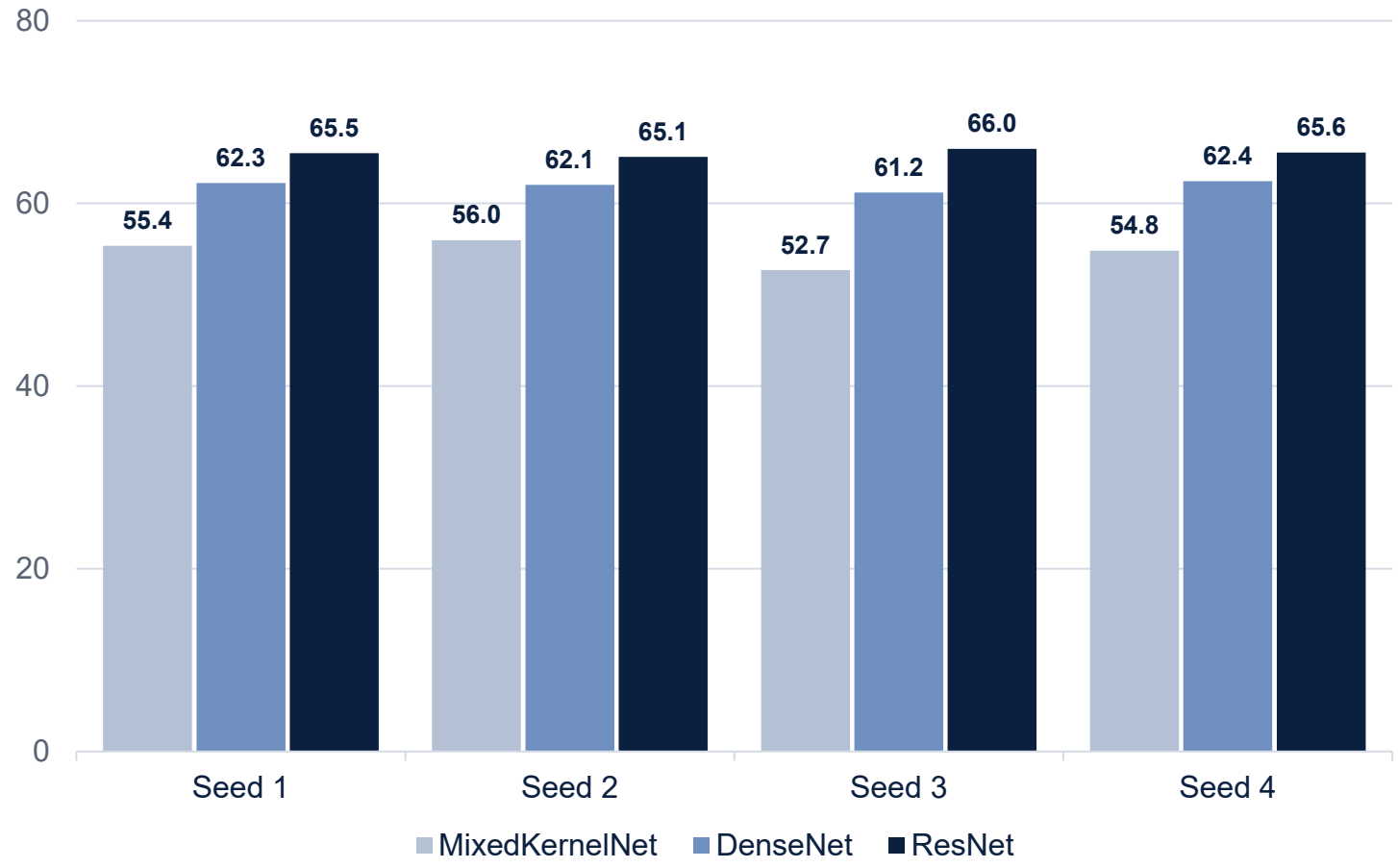
Architecture	Parameters	Design	Val κ	Test accuracy	Test κ
MixedKernelNet	1.24 M	4 conv blocks alternating 3×3 and 5×5 kernels	0.688	54.7 %	0.655
DenseNet	0.23 M	3 dense blocks (growth 16) with 1×1 transitions	0.776	62.0 %	0.752
ResNet (selected)	**5.31 M**	**4 stages of residual blocks, 64 → 256 channels**	**0.811**	**65.5 %**	**0.817**
Ensemble of all 12 models	–	κ-weighted, flip TTA, bias tuned on validation	–	64.72 %	0.796
Ensemble of 4 ResNets (final)	**–**	**κ-weighted, flip TTA, bias tuned on validation**	**–**	**65.67 %**	**0.813**

Note: single-model rows are means over seeds 1 to 4 (validation κ is the selection metric); test numbers are for reporting only. The 4-ResNet ensemble comes from a separate ResNet-only run of the same recipe; before bias tuning the two ensembles score 64.67 % / κ 0.787 (12 models) and 65.82 % / κ 0.819 (4 ResNets).

Per-Seed Accuracy by Architecture

Each of the 12 models scored on all 4,008 test images; the ranking holds for every seed

1 TEST ACCURACY (%) OF EACH MODEL



10.8 pts

higher mean accuracy for ResNet than for MixedKernelNet (65.5 % vs 54.7 %)

DenseNet sits in between at 62.0 % with only 0.23 M parameters.

64.67 %

raw accuracy of the 12-model ensemble, below the best single ResNet (65.99 %)

The final ensemble therefore keeps the four ResNets only.

Results

Four-seed ResNet ensemble, trained from scratch

65.67 %
Accuracy
all 4,008 test images

0.813
Quadratic κ
all 4,008 test images

0.665
Macro-F1
all 4,008 test images

65.94 %
Accuracy
official 1,656 test images

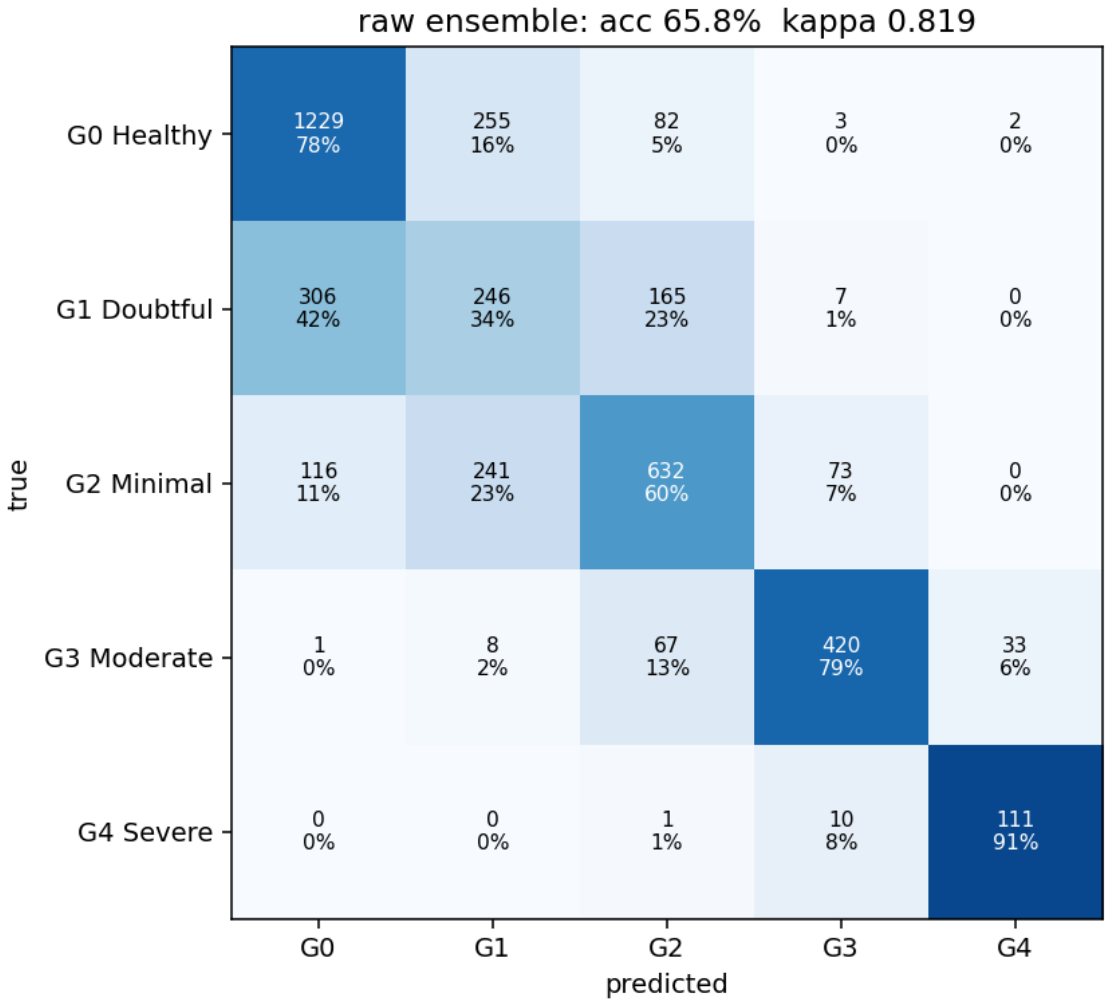
1 FULL EVALUATION (SCORED ONCE ON TEST)

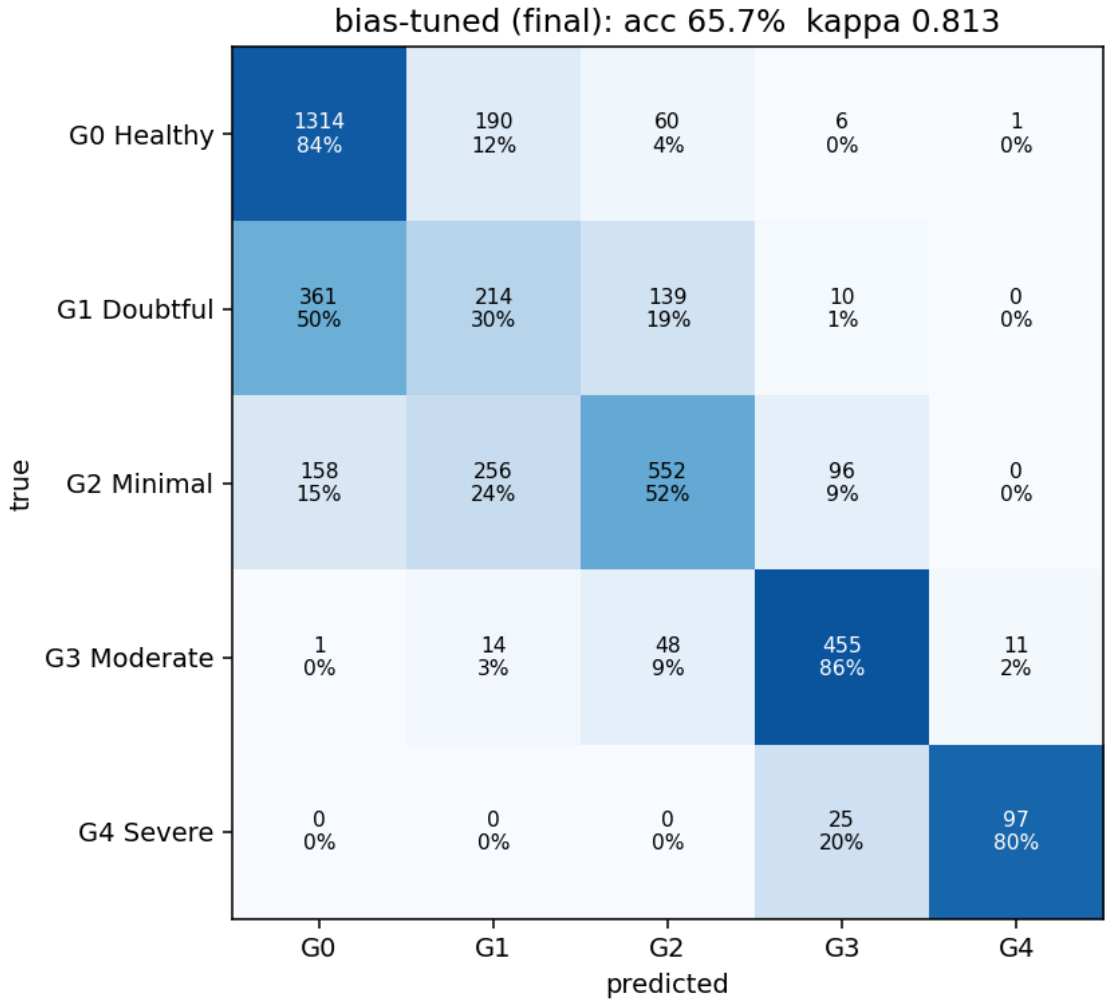
Evaluation set	Images	Accuracy	Balanced acc.	Macro-F1	Quad. κ
All test – final (bias-tuned)	4,008	65.67 %	0.661	0.665	0.813
All test – raw	4,008	65.82 %	0.684	0.672	0.819
Official test – final (bias-tuned)	1,656	65.94 %	0.657	0.661	0.825
Official test – raw	1,656	66.61 %	0.689	0.677	0.833

Final applies the per-class bias tuned on validation; **raw** is the plain κ-weighted ensemble. Bias tuning trades about 0.2 points of accuracy for better balance between grades; both are reported.

Confusion Matrices

Raw ensemble (left) and final bias-tuned ensemble (right) on all 4,008 test images; rows = true grade, columns = predicted

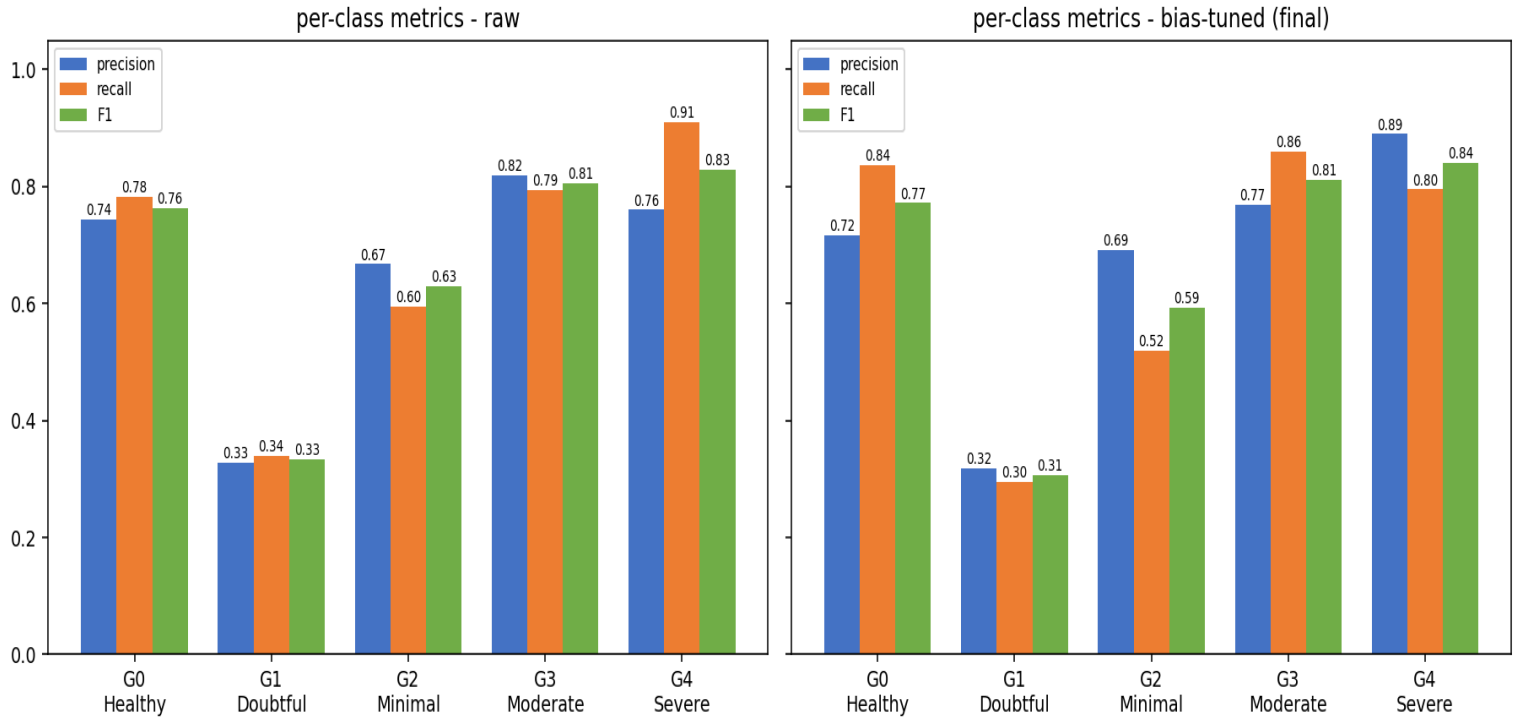




Per-Class F1

Final ensemble on all 4,008 test images; G1 (Doubtful) is the weakest grade, G3 and G4 the strongest

1 PER-CLASS F1 (FINAL ENSEMBLE)



93.8 %

of predictions are within one grade of the truth

G1 (Doubtful) is the hardest grade; it is most often read as G0 or G2.

Comparison with Published Work

Studies on the same knee-OA dataset (Chen / Kaggle), numbers as reported by the authors

Work	Model	Pretrained?	Accuracy	Other reported
Chen et al., 2019	VGG-19 + ordinal loss	Yes (fine-tuned)	69.7 %	MAE 0.344
Yong et al., 2022	DenseNet-161 + ordinal regression	Not checked	70.2 %	–
Jain et al., 2024	OsteoHRNet (multi-scale attention)	Not checked	71.7 %	–
Jeong & Lee, 2025	CLIP-KOA (CLIP ViT-B/16)	Yes (CLIP)	71.9 %	F1 0.709
Pi et al., 2023	Ensemble of 6 CNNs	Yes (ImageNet)	76.9 % *	F1 0.767
Momenpour & Abu Mallouh, 2025	EfficientNetB5 baseline	Yes (ImageNet)	66.4 %	macro-F1 0.602
afzall003 (GitHub)	Custom CNN, 1.18 M params	No (from scratch)	not reported	bal. acc. 0.644, κ 0.753
This work	**ResNet × 4 ensemble, 5.31 M params**	**No (from scratch)**	**65.9 %**	**bal. acc. 0.657, macro-F1 0.661, κ 0.825**

Test set: all rows use the 1,656-image official test split as reported, except * Pi et al., who use five-fold cross-validation on the remaining images and do not state clearly which split the headline number uses. Protocols, label cleaning and metrics differ between studies.

Our Work vs Existing Work

Approach and results side by side; existing numbers as reported by the authors

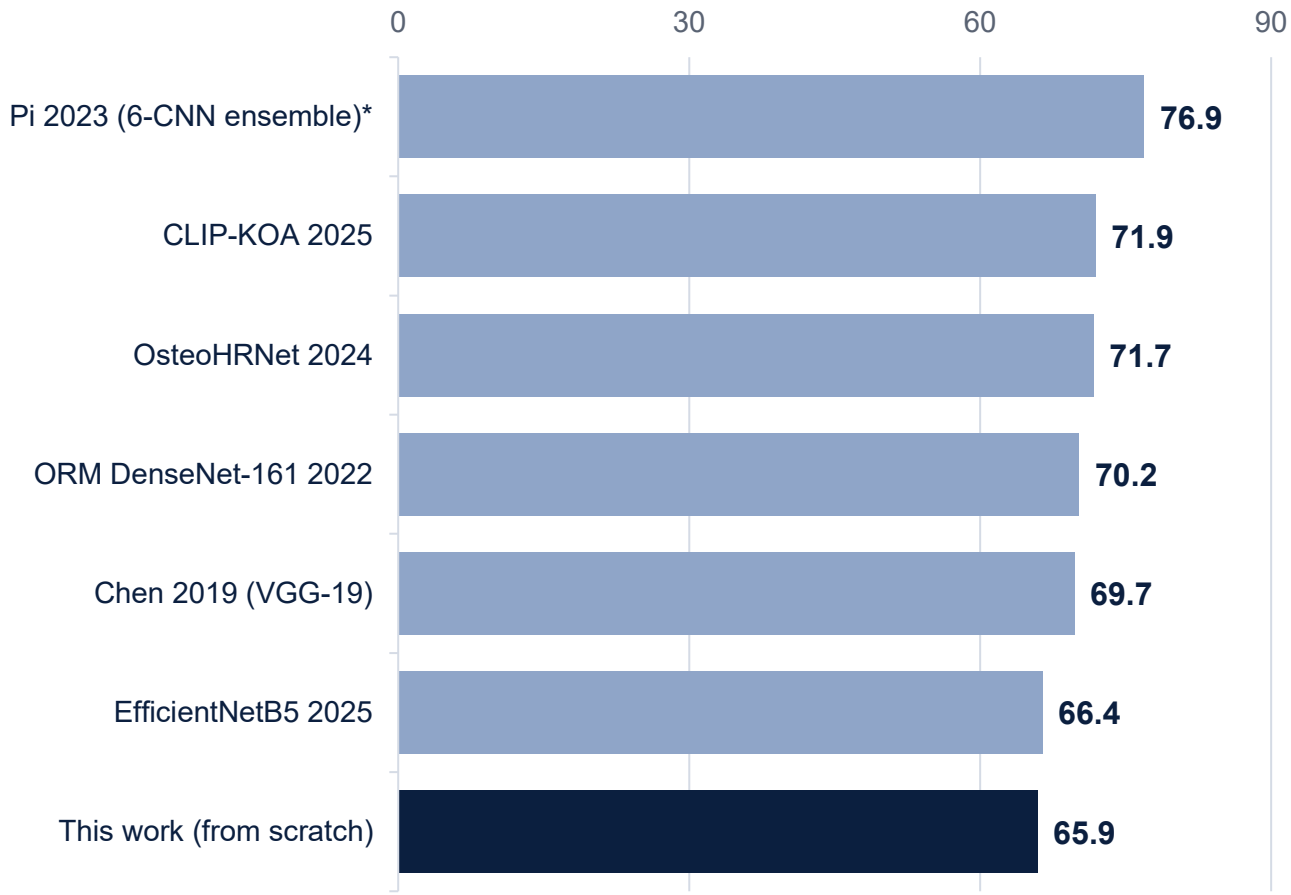
Aspect	Existing work	This work
Starting weights	4 of 7 listed works start from ImageNet or CLIP weights; 1 trains from scratch, 2 not checked	**Random initialisation; only rotation-prediction pre-training on our own training images**
Architecture	VGG-19, DenseNet-161, OsteoHRNet, CLIP ViT-B/16, EfficientNetB5; a 1.18 M-parameter CNN from scratch	**Compact 4-stage ResNet with 5.31 M parameters**
Key techniques	Ordinal loss or regression, multi-scale attention, CLIP backbone, one CNN ensemble	**CleanLab label cleaning, rotation self-supervision, focal loss with ordinal soft labels, four-seed ensemble**
Evaluation	Official 1,656-image test split as reported; Pi et al. use five-fold cross-validation	**Official 1,656 images and all 4,008; nothing selected or tuned on test**
Accuracy	66.4 – 76.9 % across the six works that report it	**65.94 % on the official split, 65.67 % on all 4,008**
Balanced acc. / κ	From-scratch CNN: 0.644 / 0.753; EfficientNetB5 macro-F1 0.602	**0.657 / 0.825; macro-F1 0.661**

Takeaway: with no pretrained weights, this work is within 0.5 point of an ImageNet EfficientNetB5 baseline and ahead of the other from-scratch CNN on balanced accuracy and κ. It stays 4 to 11 points below the best pretrained models. Protocols and metrics differ between studies.

Where This Work Stands

Accuracy on the same dataset; every other model starts from pretrained weights or was not checked

1 REPORTED ACCURACY (%)



2 WHAT THE COMPARISON SHOWS

Gap to pretrained models

4 to 11 points lower accuracy than the 69.7 – 76.9 % reported with pretrained backbones, and within 0.5 point of an ImageNet EfficientNetB5 baseline (66.4 %), whose macro-F1 is 0.602 against our 0.661.

Ahead among from-scratch models

Balanced accuracy 0.657 and κ 0.825, against 0.644 and 0.753 for the other from-scratch CNN we found on the same official test set.

A fair reading

Pretraining is a key difference: the stronger works start from ImageNet or CLIP weights, ours from random weights plus rotation pre-training.

* Pi et al.: five-fold cross-validation, split of the headline number unclear. Results on relabelled test labels or small subsets are excluded.

Conclusion & Future Work

What the project shows and where it stops

1 KEY OUTCOMES

- **Accuracy:** 65.67 % on 4,008 test images, quadratic κ 0.813
- **Official test:** 65.94 %, κ 0.825, from scratch
- **Ensemble:** four ResNet seeds, validation κ 0.815 – 0.823 each
- **Label noise:** CleanLab down-weighted 988 suspect labels without discarding data
- **Efficiency:** the whole pipeline ran in 2.7 hours on one Kaggle T4

2 LIMITATIONS

- G1 (Doubtful) has F1 of only 0.31; it is hard to separate from G0 and G2
- Bias tuning trades a little accuracy for better balance between grades
- One dataset only; no external validation
- Below the best pretrained ensembles reported in the literature
- A research prototype, not a medical device

Takeaway. A compact ResNet, cleaned labels and a four-seed ensemble reach κ 0.81 without any pretrained weights.

Future work: larger and external data, pretrained backbones as a comparison, and calibrated uncertainty for borderline grades.